

“ Sleep, Stress and Mobile Usage Data Analysis Using Correlation and Pairplots”

A PROJECT SUBMITTED TO

SYNTECXHUB Technologies

Kanpur Nagar, Uttar Pradesh

FOR THE INTERNSHIP

IN

DATA SCIENCE

By

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INTRODUCTION

In today's digital world, smartphones have become an important part of daily life. People use mobile phones for communication, entertainment, education, and many other purposes. However, excessive use of mobile devices can affect a person's health, especially sleep patterns, stress levels, and mental fatigue.

Sleep plays a very important role in maintaining both physical and mental health. But due to increased screen time, especially before sleep, many people experience poor sleep quality and shorter sleep duration. This can lead to higher stress levels and increased mental tiredness.

This project focuses on analyzing the relationship between important variables such as daily screen time, sleep duration, sleep quality, stress level, and mental fatigue. The dataset used in this project provides useful information about individuals' daily habits and lifestyle patterns.

To understand these relationships, correlation analysis is used to measure how strongly the variables are related to each other. Visualization techniques like heatmap and pairplot are used to clearly show the relationships, patterns, and trends in the data.

The main aim of this project is to understand how mobile usage and sleep habits affect stress and mental fatigue. This study helps in identifying how increasing screen time and poor sleep can negatively impact mental health.

Overall, this project highlights the importance of maintaining a balance between screen time and sleep for better mental well-being. It also shows how data analysis can be used to understand real-life problems and provide meaningful insights.

OBJECTIVES

- To load and select important numerical variables using Pandas.
- To find the correlation between numerical variables and visualize it using a heatmap.
- To analyze pairwise relationships using pairplot.
- To identify strong positive and negative relationships between variables

METHODOLOGY

Data Source:

The dataset used for this project is the Sleep, Mobile Usage and Stress Dataset, obtained from Kaggle. It contains 15,000 rows and 13 columns with information about screen time, sleep patterns, stress levels, and mental fatigue etc.

Variables Description:

User ID: Unique identification number of each user

- Age: Age of the individual
- Gender: Gender of the individual (Male/Female)
- Occupation: Profession of the individual
- Daily Screen Time Hours: Time spent on mobile devices per day
- Phone Usage Before Sleep Minutes: Mobile usage time before going to sleep
- Sleep Duration Hours: Number of hours a person sleeps
- Sleep Quality Score: Quality of sleep (higher value indicates better sleep)
- Stress Level: Level of stress experienced by the individual
- Caffeine Intake Cups: Number of caffeine drinks consumed per day
- Physical Activity Minutes: Time spent on physical activities per day
- Notifications Received Per Day: Number of notifications received daily
- Mental Fatigue Score: Level of mental tiredness or exhaustion

Process:

First, the dataset was loaded using the Pandas library and basic checks were performed to understand its structure and data types. Then, important numerical variables such as screen time, sleep duration, sleep quality, stress level, and mental fatigue were selected for analysis.

Next, the correlation between these variables was calculated and visualized using a heatmap.pairplot was also created to study the relationships and patterns between variables. Finally, the strongest positive and negative correlations were identified and interpreted to draw meaningful conclusions about the impact of mobile usage and sleep on stress and mental fatigue.

DATA ANALYSIS

❖ To load and select important numerical variables using Pandas.

Code:

```
import pandas as pd
# Load dataset
df = pd.read_csv("C:\\Users\\HP\\OneDrive\\Scans\\Scans\\RESEACH PROJECT
PAPER\\sleep_mobile_stress_dataset_15000.csv")

# Dataset info
print("\nDataset Info:\n")
df.info()

# Select only useful numeric columns
df_selected = df[[
    'age',
    'daily_screen_time_hours',
    'phone_usage_before_sleep_minutes',
    'sleep_duration_hours',
    'sleep_quality_score',
    'stress_level',
    'caffeine_intake_cups',
    'physical_activity_minutes',
    'notifications_received_per_day',
    'mental_fatigue_score'
]]
print("\nSelected Columns:\n")
print(df_selected.head())
```

Output:

Dataset Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 15000 entries, 0 to 14999

Data columns (total 13 columns):

	Column	Non-Null Count	Dtype
0	user_id	15000 non-null	int64
1	Age	15000 non-null	int64
2	Gender	15000 non-null	object
3	Occupation	15000 non-null	object
4	daily_screen_time_hours	15000 non-null	float64
5	phone_usage_before_sleep_minutes	15000 non-null	int64
6	sleep_duration_hours	15000 non-null	float64
7	sleep_quality_score	15000 non-null	float64
8	stress_level	15000 non-null	float64
9	caffeine_intake_cups	15000 non-null	int64
10	physical_activity_minutes	15000 non-null	int64
11	notifications_received_per_day	15000 non-null	int64
12	mental_fatigue_score	15000 non-null	float64

dtypes: float64(5), int64(6), object(2)

memory usage: 1.5+ MB

Selected Columns:

age	daily_screen_time_hrs	phone_usage_before_sleep_min	sleep_duration_hrs	sleep_quality_score	stress_level
56	3.26	86	5.31	7.72	3.49
46	1.8	32	7.36	9.70	3.01
32	3.4	107	4.50	6.38	5.03
25	9.00	36	6.68	5.53	10.00
38	3.52	56	7.57	6.69	6.71

caffeine_intake_cups	physical_activity_min	notifications_received_per_day	mental_fatigue_score
0	35	119	3.57
0	16	299	1.91
0	17	21	6.05
0	3	220	9.92
4	92	167	5.99

❖ To find the correlation between numerical variables and visualize it using a heatmap.

Code:

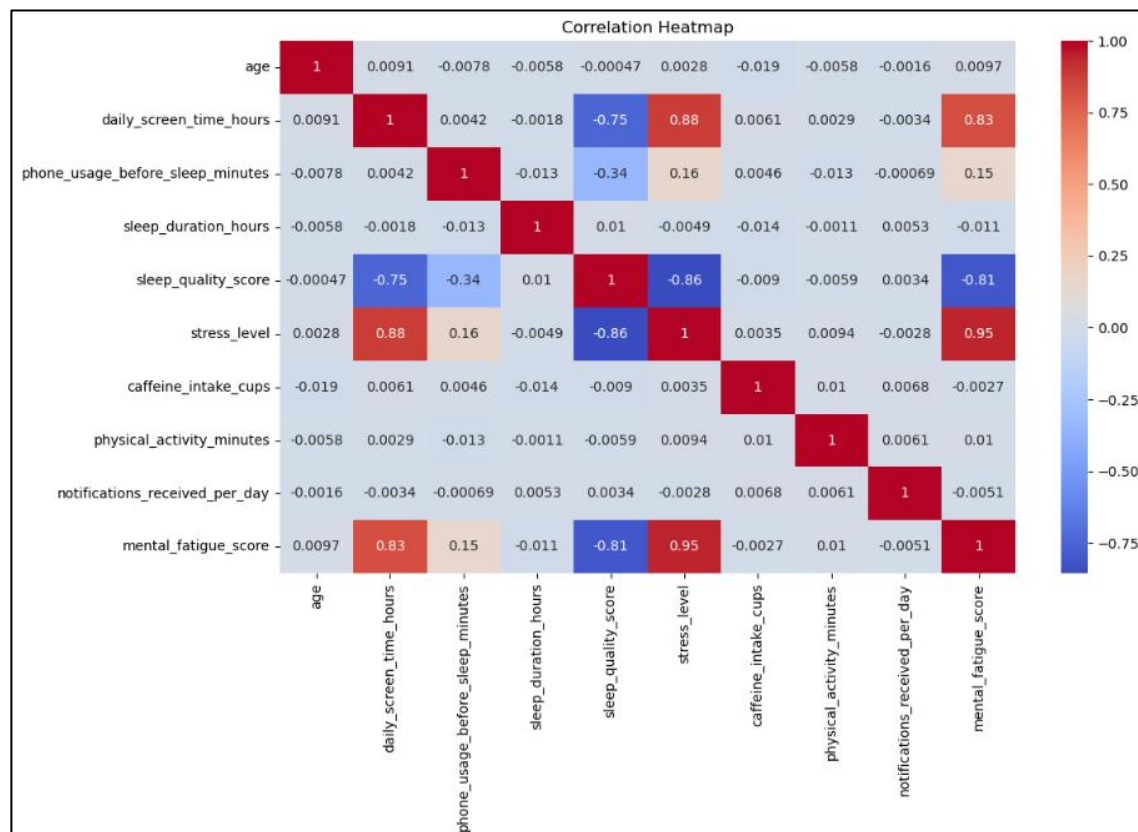
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv("C:\\Users\\HP\\OneDrive\\Scans\\Scans\\RESEACH PROJECT
PAPER\\sleep_mobile_stress_dataset_15000.csv")
# Correlation matrix
correlation = df_selected.corr()
print("\nCorrelation Matrix:\n")
print(correlation)
plt.figure(figsize=(12,7))
sns.heatmap(correlation, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

Output:

Correlation Matrix:

	age	daily_scre n time_hrs	phone_usage _ before_sleep _min	sleep_ duration_hr s	sleep_ quality_scor e	stress_leve l
age	1.000	0.0090	-0.0077	-0.0058	-0.000466	0.002849
daily_screen_ time_hrs	0.000	1.0000	0.0042	-0.00183	-0.746531	0.879001
phone_usage_ before_sleep min	-0.000	0.0004	1.0000	-0.012930	-0.335283	0.155330
sleep_ duration_hrs	-0.004	-0.0018	-0.0129	1.000000	0.009990	-0.004893
sleep_ quality_score	0.002 0	-0.7465	-0.3352	0.009990	1.000000	-0.857463
stress_level	-0.019	-0.8790	0.1553	-0.004893	-0.857463	0.000000
caffeine_intake_ cups	-0.019	0.0061	0.0046	-0.01	1.000000	1.000000
physical_activity_ min	-0.005	0.0028	-0.01127	-0.0011	0.003458	0.010249
notifications_received_per_d ay	-0.001	-0.0033	-0.0006	0.0053	0.009360	0.006762
mental_fatigue_score	0.009	0.8285	0.1501	-0.0110	0.948768	-0.002722

	caffeine_intake_cups	physical_activity_min	notifications_received_per_day	mental_fatigue_score
age	-0.01918	-0.00582	-0.001594	0.009724
daily_screen_time_hrs	0.006128	0.002899	-0.003357	0.828368
phone_usage_before_sleep_min	0.04639	-0.01270	-0.000691	0.150180
sleep_duration_hrs	-0.01368	-0.00111	0.005343	-0.011073
sleep_quality_score	-0.00903	-0.00585	-0.005857	-0.808019
stress_level	0.00345	0.009360	0.010249	-0.002808
caffeine_intake_cups	1.00000	0.010249	0.006762	-0.002722
physical_activity_min	0.01249	1.000000	0.006069	0.010254
notifications_received_per_day	0.00676	0.006069	1.000000	-0.005077
mental_fatigue_score	-0.00272	0.010254	-0.005077	1.000000



Interpretation:

The heatmap shows that stress and mental fatigue have a strong positive correlation, meaning both increase together. Screen time is positively related to stress and fatigue, while sleep quality is negatively related to both. Other variables show weak or no significant relationships.

Conclusion:

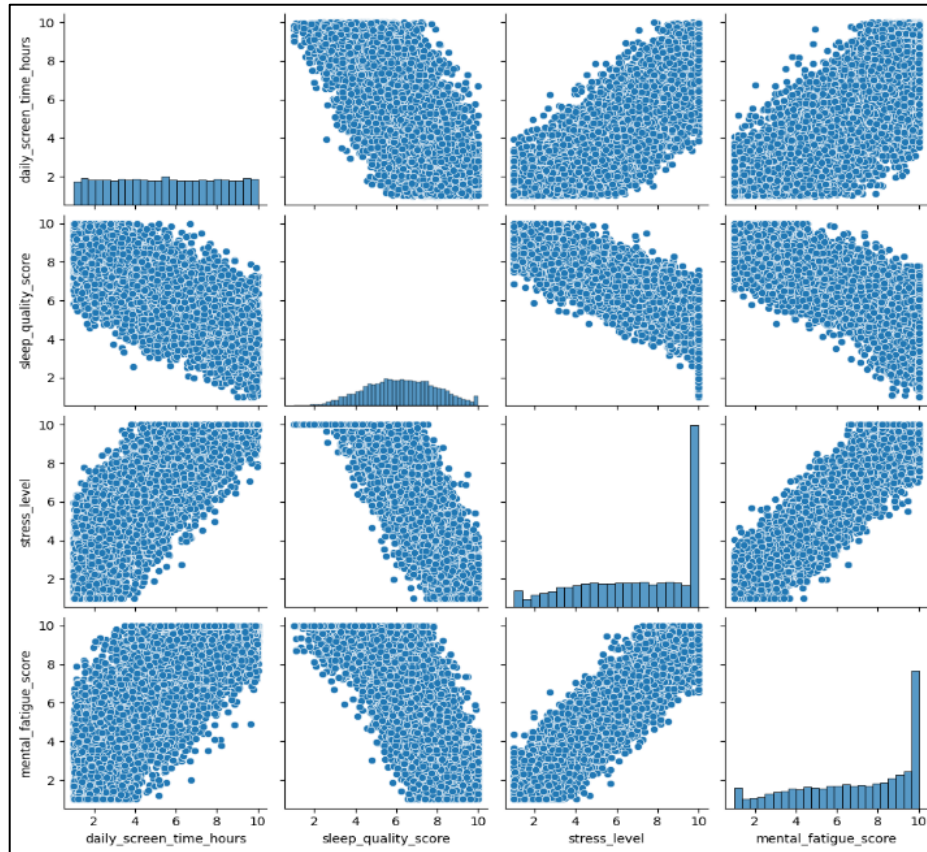
Higher screen time leads to increased stress and mental fatigue, while better sleep quality helps reduce both. Overall, maintaining good sleep habits and limiting screen usage can improve mental well-being.

❖ **To analyze pairwise relationships using pairplot.**

Code :

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv("C:\\Users\\HP\\OneDrive\\Scans\\Scans\\RESEACH PROJECT
PAPER\\sleep_mobile_stress_dataset_15000.csv")
# Select ONLY important variables
# Select ONLY important variables
df_selected = df[[
    'daily_screen_time_hours',
    'sleep_quality_score',
    'stress_level',
    'mental_fatigue_score'
]]
# Correlation
correlation = df_selected.corr()
# Pairplot (clean & clear)
sns.pairplot(df_selected, diag_kind='hist')
plt.show()
```

Output:



Interpretation:

The pairplot shows that daily screen time and mental fatigue score follow an upward trend, indicating a positive relationship. Stress level also increases with higher screen time, showing another positive association.

On the other hand, sleep quality score decreases as stress and mental fatigue increase, indicating a negative relationship between these variables.

Conclusion:

Higher screen time leads to increased stress and mental fatigue, while better sleep quality helps in reducing both. Stress and mental fatigue are strongly related, and maintaining healthy sleep habits along with reduced screen exposure can improve overall mental well-being.

❖ **To identify strong positive and negative relationships between variables.**

Code:

```
# Unstack correlation matrix
corr_pairs = correlation.unstack()

# Remove self correlation
corr_pairs = corr_pairs[corr_pairs != 1]

# Sort values
sorted_pairs = corr_pairs.sort_values()
print("Top Negative Correlations:\n")
print(sorted_pairs.head(5))
print("\nTop Positive Correlations:\n")
print(sorted_pairs.tail(5))
```

Output:

Strongest Negative Correlations:

stress_level	sleep_quality_score	-0.857463
mental_fatigue_score	sleep_quality_score	-0.808019
sleep_quality_score	daily_screen_time_hours	-0.746531

Strongest Positive Correlations:

daily_screen_time_hours	mental_fatigue_score	0.828368
stress_level	daily_screen_time_hours	0.879001
mental_fatigue_score	stress_level	0.948768

Conclusion:

The strongest positive relationship is between stress level and mental fatigue score (0.95), meaning higher stress increases fatigue.

The strongest negative relationship is between sleep quality score and stress level (-0.86), showing better sleep reduces stress.

Overall, higher screen time increases stress and fatigue, while good sleep quality helps reduce them.

❖ **Overall Conclusion:**

This project analyzed the dataset to understand how key variables like daily screen time, sleep quality, stress level, and mental fatigue are related. The analysis shows that stress level has the strongest positive impact on mental fatigue, meaning higher stress leads to greater mental exhaustion. Screen time also has a positive effect on both stress and fatigue, indicating that increased mobile usage increases mental pressure.

On the other hand, sleep quality has a negative effect on stress and mental fatigue, showing that better sleep helps in reducing both. Sleep duration, however, shows a weaker relationship compared to sleep quality.

From a real-life perspective, the results suggest that individuals should focus on maintaining good sleep quality and limiting excessive screen time. Simply spending more time on mobile devices can increase stress and fatigue, while proper sleep habits can improve overall mental well-being. Therefore, maintaining a balance between screen time and sleep is important for a healthy lifestyle.